

Structural skeleton of $B_{\ell+1}^{(\lambda)} V_\lambda$: the leftmost-and-rightmost-factor theorem

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Abstract

The May-13 morning paper refuted the naive Hecke-iso upgrade of the dim-formula $\dim B_{\ell+1}^{(\lambda)} V_\lambda = f^{\lambda^{(\geq 2)}} : \text{no parabolic embedding } H_q(S_{n-\ell}) \hookrightarrow H_q(S_n) \text{ acts on } B$. Computationally, the only generator that *structurally* acts as a scalar on B was T_ℓ (acting as q), with shape-dependent additions $T_1, \dots, T_{k(\lambda)}$ acting as -1 .

We prove the two universal facts here, both with one-line proofs from the Hecke relations:

Theorem A. $T_\ell \cdot \Omega = q \cdot \Omega$ in H_q , where $\Omega := R'_{\ell+1} R'_{\ell+2} \cdots R'_n$. Consequently T_ℓ acts as scalar q on $B_{\ell+1}^{(\lambda)} V_\lambda$, i.e. $B_{\ell+1}^{(\lambda)} V_\lambda \subseteq E_\ell^+|_{V_\lambda}$.

Theorem A'. $\Omega \cdot T_1 = q \cdot \Omega$ in H_q . Consequently Ω annihilates the (-1) -eigenspace $E_1^- = \ker(T_1 + 1)|_{V_\lambda}$, and the induced map $V_\lambda/E_1^- \rightarrow E_\ell^+$ has image $B_{\ell+1}^{(\lambda)} V_\lambda$.

The two theorems express the same Hecke identity $(T_j - q)(T_j + 1) = 0$ applied at the *leftmost* and *rightmost* factor of the chain product Ω . They are dual structural skeletons of Ω and, together, the only *universal* eigenspace constraints on $B_{\ell+1}^{(\lambda)} V_\lambda$: everything else (e.g. which T_j , $j < \ell$, acts as -1) is shape-dependent.

The result tightens but does not contradict the May-13 morning refutation: T_ℓ acting as scalar q is the strongest *shape-uniform* Hecke-equivariance statement available, and we explain precisely why no analogous statement holds for T_j with $j \notin \{\ell\}$ or $j \notin \{1, \dots, k(\lambda)\}$.

Verified mod $P = 100\,003$ at $q = 11$ for $\lambda \in \{(3, 2, 1^3), (4, 2, 1^4), (3, 3, 1^4), (4, 3, 1^3), (3, 2, 1^4), (2, 1^3), (3, 1^3), (3, 2)\}$.

1 Setup

We retain the notation of the May-13 multi-corner dimension paper. $H_q = H_q(S_n)$ is the Iwahori–Hecke algebra with generators $T_1, \dots, T_{n-1}$ and relations

$$T_j^2 = (q - 1)T_j + q, \quad T_j T_{j+1} T_j = T_{j+1} T_j T_{j+1}, \quad T_i T_j = T_j T_i \quad (|i - j| \geq 2). \quad (1)$$

For $j \geq 1$, set

$$S_j := T_j + 1 \in H_q. \quad (2)$$

The Hecke relation (1) factors as $(T_j - q)(T_j + 1) = 0$. Equivalently

$$T_j S_j = q S_j, \quad S_j T_j = q S_j. \quad (3)$$

That is: S_j *absorbs* T_j from either side, producing scalar q .

Let V_λ be the Specht module in the Hoefsmit seminormal basis, indexed by SYTs of λ . Eigenspaces of T_j in V_λ :

$$E_j^+ := \ker(T_j - q)|_{V_\lambda}, \quad E_j^- := \ker(T_j + 1)|_{V_\lambda}.$$

Note $E_j^+ = \text{Im}(S_j|_{V_\lambda})$ and $E_j^- = \ker(S_j|_{V_\lambda})$.

For $k \geq 1$, define the level- k descent operator

$$R'_k := S_{k-1}S_{k-2} \cdots S_2S_1 = (T_{k-1} + 1)(T_{k-2} + 1) \cdots (T_2 + 1)(T_1 + 1) \in H_q.$$

For $\lambda \vdash n$ and $0 \leq \ell \leq n - 1$, set

$$\Omega = \Omega_\ell^{(\lambda)} := R'_{\ell+1}R'_{\ell+2} \cdots R'_n \in H_q, \quad (4)$$

the chain product from level $\ell + 1$ down to level n . We are interested in the iterated image

$$B = B_{\ell+1}^{(\lambda)} V_\lambda := \Omega \cdot V_\lambda \subseteq V_\lambda.$$

Throughout, ℓ is the length of λ (so Ω takes the “decoration-shape” product), but the structural theorems do not require this. Set $B_{\ell+1}^{(\lambda)} V_\lambda := V_\lambda$ if $\ell + 1 > n$ (empty product).

2 The two structural skeleton theorems

The Hecke absorption identity (3) drives both results. Read (4) from left to right: the *leftmost* factor of Ω is S_ℓ (the leading S in $R'_{\ell+1}$). The *rightmost* factor of Ω is S_1 (the trailing S in R'_n).

Theorem 1 (A: T_ℓ acts as scalar q). *For any partition $\lambda \vdash n$ and any ℓ with $1 \leq \ell \leq n - 1$,*

$$T_\ell \cdot \Omega = q \cdot \Omega \quad \text{in } H_q.$$

In particular, T_ℓ acts as scalar q on $B_{\ell+1}^{(\lambda)} V_\lambda$, i.e. $B_{\ell+1}^{(\lambda)} V_\lambda \subseteq E_\ell^+|_{V_\lambda}$.

Proof. The leftmost factor of $R'_{\ell+1}$ is S_ℓ : $R'_{\ell+1} = S_\ell \cdot (S_{\ell-1} \cdots S_1) = S_\ell \cdot R'_\ell$. Hence

$$\Omega = R'_{\ell+1} \cdot (R'_{\ell+2} \cdots R'_n) = S_\ell \cdot R'_\ell \cdot (R'_{\ell+2} \cdots R'_n) = S_\ell \cdot X$$

for $X := R'_\ell \cdot R'_{\ell+2} \cdots R'_n \in H_q$. By (3), $T_\ell \Omega = T_\ell S_\ell X = q S_\ell X = q \Omega$. Applying to any $v \in V_\lambda$ and writing $b = \Omega v \in B$: $T_\ell b = T_\ell \Omega v = q \Omega v = qb$. $\square$

Theorem 2 (A': Ω annihilates E_1^-). *For any partition $\lambda \vdash n$ and any ℓ with $0 \leq \ell \leq n - 1$,*

$$\Omega \cdot T_1 = q \cdot \Omega \quad \text{in } H_q.$$

Equivalently, Ω annihilates $E_1^- = \ker(T_1 + 1)|_{V_\lambda}$, and the induced map $\bar{\Omega} : V_\lambda / E_1^- \rightarrow V_\lambda$ has image $B_{\ell+1}^{(\lambda)} V_\lambda$.

Proof. The rightmost factor of R'_n is S_1 : $R'_n = (S_{n-1} \cdots S_2) \cdot S_1$. Hence

$$\Omega = (R'_{\ell+1} \cdots R'_{n-1}) \cdot R'_n = (R'_{\ell+1} \cdots R'_{n-1}) \cdot (S_{n-1} \cdots S_2) \cdot S_1 = Y \cdot S_1$$

for $Y \in H_q$. By (3), $\Omega T_1 = Y S_1 T_1 = q Y S_1 = q \Omega$.

For the equivalence: for any $v \in V_\lambda$, write Ωv . If $v \in E_1^-$, then $T_1 v = -v$, so $S_1 v = (T_1 + 1)v = 0$, hence $\Omega v = Y S_1 v = 0$.

Conversely, the identity $\Omega(T_1 v) = q \Omega(v)$ holds for all v ; if $v \in E_1^-$ then $\Omega(T_1 v) = \Omega(-v) = -\Omega(v)$, so $(-1)\Omega(v) = q \Omega(v)$, forcing $(q + 1)\Omega(v) = 0$. At generic q (in particular at $q = 11 \bmod P = 100\,003$, $q + 1 = 12 \neq 0$), this gives $\Omega(v) = 0$. Both directions show $E_1^- \subseteq \ker(\Omega|_{V_\lambda})$. $\square$

Corollary 3 (The structural skeleton). *The operator $\Omega|_{V_\lambda} : V_\lambda \rightarrow V_\lambda$ induces a linear map*

$$\bar{\Omega} : V_\lambda/E_1^- \longrightarrow E_\ell^+$$

whose image is $B_{\ell+1}^{(\lambda)} V_\lambda$. In symbols,

$$B_{\ell+1}^{(\lambda)} V_\lambda \subseteq E_\ell^+, \quad \ker(\Omega|_{V_\lambda}) \supseteq E_1^-.$$

Proof. Theorem 1 gives the codomain E_ℓ^+ ; Theorem 2 gives the kernel containment. $\square$ $\square$

Remark 4 (“Skeleton” interpretation). The two theorems are dual structural facts: the leftmost factor of Ω forces T_ℓ -eigenspace constraints *on the image* B ; the rightmost factor forces T_1 -eigenspace constraints *on the kernel*. They sit at the two “ends” of the chain product $\Omega = R'_{\ell+1} R'_{\ell+2} \cdots R'_n$ and are the only constraints from H_q alone (without using the specific representation V_λ).

3 What is NOT universal: shape-dependence

The May-13 morning refutation paper found computationally that for $\lambda = (3, 2, 1^3)$ at $n = 8$, $\ell = 5$, $\dim B = 2$:

- T_5 acts as scalar q on B (Theorem 1);
- T_1 acts as scalar -1 on B (shape-dependent);
- T_j for $j \in \{2, 3, 4, 6, 7\}$ does *not* even preserve B .

For $\lambda = (4, 3, 1^3)$, the same refutation paper showed T_1 does *not* act as -1 : there is no length- k prefix of $\{T_1, \dots, T_{\ell-1}\}$ acting as scalars beyond T_ℓ itself.

Theorem 1 explains the universal $T_\ell = q$ behaviour structurally. What about the empirical $T_1, T_2, \dots, T_k$ acting as -1 on B , for some shape-dependent $k = k(\lambda) \in \{0, 1, 2\}$?

Observation 5 (Why $T_1 = -1$ is *not* universal). The natural attempt to prove $T_1 \cdot \Omega \subseteq -1 \cdot \Omega$ fails because T_1 does *not* satisfy a left-absorption identity with Ω : T_1 commutes with T_j for $j \geq 3$ but not with T_2 , and T_2 appears (as S_2) in every R'_k for $k \geq 3$. So T_1 cannot be pulled across Ω to a position where the Hecke relation gives a scalar.

What *is* structurally true is the *right-action* identity: $\Omega T_1 = q\Omega$ (Theorem 2). This says Ω acts as zero on E_1^- (in the *source*), not that B lies in E_1^- (in the *target*). The image $\Omega(E_1^+) = B$ may or may not lie in E_1^- , depending on whether the operator $\Omega|_{E_1^+} : E_1^+ \rightarrow V_\lambda$ happens to factor through E_1^- . That is a shape-dependent condition. Empirically it holds for $\lambda = (3, 2, 1^3)$ and many other shapes; for $\lambda = (4, 3, 1^3)$ it does not.

The structural skeleton (Corollary 3) is the strongest *shape-uniform* statement available. Everything beyond it is shape-dependent, and that is precisely what the May-13 morning refutation paper made explicit.

4 Generalisation to all levels of the chain

The same proof of Theorem 1 applies at every level of the chain product, by replacing ℓ with any j in the construction. Define

$$B_{j+1} = B_{j+1}^{(\lambda)} V_\lambda := R'_{j+1} R'_{j+2} \cdots R'_n \cdot V_\lambda, \quad 0 \leq j \leq n-1.$$

(So B_{j+1} is the chain product starting from level $j+1$.)

Theorem 6 (Level- j scalar fact). *For every $j \in \{1, 2, \dots, n-1\}$ and every $\lambda \vdash n$:*

$$T_j \cdot (R'_{j+1} R'_{j+2} \cdots R'_n) = q \cdot (R'_{j+1} R'_{j+2} \cdots R'_n) \quad \text{in } H_q.$$

Hence T_j acts as scalar q on $B_{j+1}^{(\lambda)} V_\lambda$, i.e. $B_{j+1}^{(\lambda)} V_\lambda \subseteq E_j^+|_{V_\lambda}$.

Proof. Same as Theorem 1, with ℓ replaced by j . □ □

Corollary 7 (Eigenspace chain). *The descending chain $B_n \supseteq B_{n-1} \supseteq \cdots \supseteq B_{\ell+1}$ need not be literally nested (each B_{j+1} is the image of R'_{j+1} on the next), but each B_{j+1} is contained in E_j^+ :*

$$B_{j+1} \subseteq E_j^+ \quad \text{for all } 1 \leq j \leq n-1.$$

Remark 8. This sharpens the May-13 morning Step-1 conjecture (PROVE.MD, “ B is fixed under $(T_{j-1}+1)$ for $j > \ell$ ”). The correct universal statement is: B_{j+1} is in the $+q$ -eigenspace of T_j , not in any joint eigenspace involving $T_{j+1}, T_{j+2}, \dots$. The latter would have been the Hecke-iso statement that was refuted.

5 Computational verification

We verify Theorem 1 (specifically, $T_\ell \cdot W = q \cdot W$ where W is an explicit column basis of B) and Theorem 2 (specifically, $\Omega \cdot E_1^- = 0$ as a matrix equation) mod $P = 100\,003$ at $q = 11$, using the Hoefsmit seminormal matrix model (script `compute_dim.B.py` of May-13).

λ	n	ℓ	$\dim V_\lambda$	$\dim B$	$T_\ell = q$ on B ?	$\Omega(E_1^-) = 0$?
$(3, 2)$	5	2	5	2	yes	yes
$(2, 1, 1, 1)$	5	4	4	1	yes	yes
$(3, 1, 1, 1)$	6	4	10	1	yes	yes
$(3, 2, 1^3)$	8	5	64	2	yes	yes
$(3, 2, 1^4)$	9	6	105	2	yes	yes
$(4, 2, 1^4)$	10	6	350	3	yes	yes
$(3, 3, 1^4)$	10	6	225	2	yes	yes
$(4, 3, 1^3)$	10	5	525	5	yes	yes

(Scripts: 2026-05-13-evening/verify_Tell_scalar.py, 2026-05-13-evening/verify_omega_kills_E1minus.)

In all 8 shapes, both structural facts hold, and the $\dim B$ column matches the May-13 morning closed form $f^{\lambda^{(\geq 2)}}$ where applicable. The proofs of Theorems 1 and 2 are themselves Hecke-algebra identities, so they hold over $\mathbb{Z}[q]$ uniformly; the computational table is a sanity check, not part of the proof.

6 What this gives the iso programme

The May-13 morning refutation paper identified three resolutions for the dim-equality $\dim B = f^{\lambda^{(\geq 2)}}$:

- (a) iso via exotic operators in H_q ;
- (b) vector-space-only iso (combinatorial bijection);

(c) subquotient or induced Hecke structure.

The structural skeleton (Cor. 3) constrains resolution (a): any exotic operator $X \in H_q$ that intertwines an $H_q(S_{n-\ell})$ -action on B with $V_{\lambda(\geq 2)}$ must commute with T_ℓ (so as to preserve the $T_\ell = q$ eigenspace property of B). Hence X lies in the centraliser $Z_{H_q}(T_\ell)$, which is generated by $\{T_j : j \neq \ell-1, \ell+1\} \cup \{T_\ell\}$ as a Hecke-style subalgebra of type $A_{\ell-2} \times A_{n-\ell-2}$, with T_ℓ acting as scalar q . However the refutation paper showed (computationally) that $T_1, \dots, T_{\ell-2}$ and $T_{\ell+2}, \dots, T_{n-1}$ generally do *not* preserve B as a subspace. So even within $Z_{H_q}(T_\ell)$, the generic generators fail. Resolution (a) requires *non-trivial* elements of $Z_{H_q}(T_\ell)$, beyond the simple generators.

A natural candidate is the family of Jucys–Murphy elements

$$L_k = T_{k-1} \cdots T_2 T_1 \cdot T_1 T_2 \cdots T_{k-1}, \quad 1 \leq k \leq n,$$

which generate a maximal commutative subalgebra of H_q . The L_k commute pairwise; L_k commutes with T_j when $j \notin \{k-1, k\}$. In particular L_k commutes with T_ℓ when $k \notin \{\ell, \ell+1\}$. We have not investigated whether such L_k preserve B as a subspace, and leave this as an open direction for the iso programme.

The dual structural fact (Theorem 2: $\Omega(E_1^-) = 0$) gives the parallel constraint on the *source*: any natural H_q -equivariant lift $\tilde{B} \rightarrow B$ (e.g. from an induced representation) must factor through the quotient V_λ/E_1^- .

7 Status and open directions

Proved (this paper). Theorems 1, 2, 6, with one-line proofs from the Hecke identity $(T_j - q)(T_j + 1) = 0$. These constitute the universal structural skeleton of $B_{\ell+1}^{(\lambda)} V_\lambda$.

Already known (May-13 morning).

- Dim formula $\dim B = f^{\lambda(\geq 2)}$ in the fully-stable regime (multi-corner dimension paper).
- Refutation of naive parabolic Hecke iso (decoration-iso-failed paper).
- Empirical $T_1, \dots, T_{k(\lambda)}$ act as -1 on B , where $k(\lambda) \in \{0, 1, 2\}$ is shape-dependent.

Open.

- Characterise the shape-dependence of $k(\lambda)$ (when does T_j act as -1 on B , for $j < \ell$?).
- Identify the precise stability threshold $q_0(\hat{\lambda})$ for $\dim B = f^{\lambda(\geq 2)}$ (non-monotone in q in general).
- Investigate Jucys–Murphy elements L_k with $k \notin \{\ell, \ell+1\}$: do any commute with the projector to B ? This is the natural setting for resolution (a) of the iso programme.
- Express B as a subquotient / induced module (resolution (c)).

Honest assessment. Theorems 1 and 2 are not deep. They are one-line consequences of the basic Hecke identity. Their value is in *naming* the structural skeleton precisely and explaining why no analogous statement holds for other T_j . Combined with the May-13 morning refutation, the rep-theoretic story of $B_{\ell+1}^{(\lambda)} V_\lambda$ now has a clean “universal core + shape-dependent decoration” structure:

- Universal core: $B \subseteq E_\ell^+, \ker(\Omega) \supseteq E_1^-$.

- Shape-dependent decoration: which of $T_1, \dots, T_{\ell-1}$ acts as -1 on B ; the stability threshold $q_0(\widehat{\lambda})$; the precise dimension when not in the stable regime.